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Century-long data reveals complex trends in ice cover in the Laurentian Great Lakes

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Title: Century-long data reveals complex trends in ice cover in the Laurentian Great Lakes

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Abstract:
Ice cover on large lakes regulates regional weather and aquatic ecosystems, and understanding lake ice responses to climate change is critical for environmental management. However, lake-wide ice records are generally restricted to the remote sensing era, limiting inferences about long-term shifts in climate. Here, we use continuous air temperature records to reconstruct lake-wide ice cover for each of the Laurentian Great Lakes between 1898 and 2022, assessing the skill of a statistical model against satellite observations from recent decades. Reconstructed ice records reveal a prominent multidecadal oscillation (~60-year period) superimposed on a century-scale warming trend that is significantly weaker than trends inferred over recent decades (1975 - 2022). We find that trends and variability are strongly associated with the Atlantic Multidecadal Oscillation, indicating a persistent ocean-atmosphere-lake interaction that modulates North American winter conditions. Results show that short observational windows can overstate ice trends and underrepresent internal variability. These findings highlight the need for century-scale records to disentangle decadal climate variability from the persistent effects of anthropogenic warming when assessing regional cryosphere change.

Key Points:
(1) Air temperature observations can be used to estimate lake-wide ice cover in the Laurentian Great Lakes, providing a method for reconstructing historical ice records in the region
(2) Observed air temperature and derived ice cover records in the region are characterized by a distinct multidecadal oscillation (period: ~60 years) superimposed on a relatively mild long-term warming trend (1898 - 2022), which is nearly an order of magnitude lower than recent trends (1975 - 2022).
(3) Climate time series highlight multidecadal shifts in variability, with a mid-century (1941 - 1975) decrease in variance for both annual average air temperature and ice cover.

1. Introduction

Seasonal ice cover in large lakes plays a major role in regulating regional weather[1], navigation[2], and aquatic ecosystems[3] across the Northern Hemisphere. Ice dynamics are also a sensitive indicator of changes in winter climate[4]. Shifts in the length and extent of the ice season propagate to lake-effect snowfall[5], spring stratification[6], and nutrient cycling[7], with significant ecological and socio-economic consequences. Interpreting long-term change, however, is challenging: climate teleconnections modulate winter severity at interannual to multidecadal scales, and the observational record generally spans only recent decades, making it difficult to separate internal variability from the underlying anthropogenic warming signal[8].

As the largest freshwater system with seasonal ice cover, the Laurentian Great Lakes have been recognized as a global "hotspot" of climate change, having experienced more rapid warming in recent decades than other large lakes and oceans worldwide[9,10]. As near-surface air temperatures in the region have risen over the last 30 - 50 years (~0.5°C/decade[11]), the lakes have experienced accelerated surface warming (~1°C/decade[12]) and ice cover declines (~10%/decade[13]). Part of this acceleration is attributed to winter ice–albedo feedbacks[11,14]; as ice cover declines, net albedo drops, and more solar radiation can be absorbed by the lakes over the winter.

While global warming signals manifest as century-scale trends, most observational climate change studies in the Great Lakes are limited to the modern remote sensing era: 1970s - present day[12,15]. Hydrodynamic simulations are similarly limited by the availability of high-resolution atmospheric forcing data, and, while some reanalysis datasets extend as far back as 1940 (e.g., ERA5), they are less accurate for earlier simulation years[16]. These relatively short time-scale datasets are problematic for quantifying long-term climate trends. For example, recent studies in the Great Lakes[8,17-19] have highlighted significant correlations between recent perceived climate trends and multidecadal climate oscillations, like the Atlantic Multidecadal Oscillation (AMO; 60 - 80 year period) and the Pacific Decadal Oscillation (PDO; 20 - 30 year period). Without century-scale climate records, it is impossible to disentangle these low-frequency oscillations from long-term warming trends, making it difficult to predict how the lakes may continue to change in the future.

In the Laurentian Great Lakes, studies of multidecadal trends are typically limited to land-based meteorological (e.g., air temperature, wind speed) and hydrological (e.g., streamflow, precipitation) variables, which have been recorded consistently since the early twentieth century[20-23]. Of these parameters, air temperature records are perhaps the most reliable and accurate long-term dataset, with daily observations going back to 1898[24]. Although multiple factors influence lake-ice characteristics, air temperature exerts an outsized influence on seasonal ice cover[25,26], providing a physically-based proxy for historic ice conditions. Assel[27] utilized this relationship to estimate daily ice cover from air temperature records in Lakes Erie and Superior between 1897–1983, demonstrating the feasibility of reconstructing historical ice variability from on-shore meteorological records. However, these efforts covered only part of the Great Lakes basin and ended in 1983, predating the satellite-era decline that underpins most recent ice-loss assessments.

In this study, we present century-scale, lake-wide reconstructions of annual average ice cover (AAIC) for all five Great Lakes over the last 125 years by statistically linking long-term air temperature records[28] with historical ice charts and evaluating skill against satellite-era observations. This work addresses two questions: (1) How do climate trends estimated from short observational windows differ from trends computed over a century-scale baseline? and (2) To what extent do multidecadal climate oscillations explain the mean state and interannual variability of Great Lakes ice cover and surface air temperature? We assess these relationships with respect to Atlantic multidecadal variability (AMO/AMV) and the Pacific Decadal Oscillation (PDO).

2. Results

2.1 Multidecadal Trends

Long-term hindcasts (1898 - 2022) of annual average ice cover (AAIC) concentrations were reconstructed using air temperature records from stations surrounding each of the Laurentian Great Lakes (Figure S1). Air temperatures were used to estimate maximum freezing degree days (FDD_{max}), maximum thawing degree days (TDD_{max}), and antecedent heat (AH) for each season, with derived variables used to fit a beta-regression model that was calibrated and validated against satellite ice cover observations between 1973 and 2022. Best-fit model coefficients, validation metrics, and time series plots can be found in the supplementary materials (Table S1, Figure S2). The model performed well for all five Great Lakes, with root mean square error (RMSE) estimates below 6.01% and coefficients of determination (R^2) above 0.89 for both the calibration (1973 - 1993; 2004 - 2021) and validation (1994 - 2003) periods (Figure S2). Model performance was poorest for Lake Erie, where AAIC tended to be overestimated in low ice years (AAIC < 10%), with biases as large as 15% in the anomalously warm winter of 1998[29]. FDD_{max} and AH were statistically significant predictors for all five lakes, while TDD_{max} was not a significant predictor of AAIC for Lake Michigan or Lake Superior. In general, validation metrics highlight strong model performance across both warm and cold years, providing confidence in hindcast simulations over the historical record.

Time series of simulated AAIC were used to calculate long-term (1898–2022) and short-term (1975–2022) trends using Theil-Sen estimators (see Methods). Ice cover decreased substantially in all five Great Lakes, with long-term AAIC trends between 0.44 and 0.79 %/decade (Figure 1). Lake Huron experienced the largest decline over the last century, with ~10% total ice loss over the entire simulation period. Short-term trends were substantially larger for more recent decades, with maximum ice cover loss exceeding 4 %/decade in Lakes Superior and Erie. All five lakes exhibited statistically significant trends at the 95% confidence level (as estimated with Mann-Kendall tests), with the exception of Lake Superior, where long-term trends were only significant at the 90% level. Decreases in AAIC were consistent with trends in annual average air temperature (AAAT) across the basin (Figure S3), which increased by approximately 0.08 (0.40) °C/decade between 1898 and 2022 (1975 and 2022).

Detrending analysis revealed nonlinear patterns in ice cover time series. Low-frequency signals extracted using singular spectrum analysis (SSA; see Methods) showed multi-decadal variability superimposed on the long-term warming trends (Figures 1,2). Oscillation periods were consistent between AAAT and AAIC (approximately 60–80 years), and the two variables exhibited an out-of-phase relationship (i.e., AAAT maxima coinciding with AAIC minima). Signal phase differed among lakes, with peaks occurring earliest in Lakes Michigan and Superior, followed sequentially by Erie, Huron, and Ontario. Oscillation amplitudes (Figure 2c,d) were largest in Lake Erie, reflecting its shallow depth and short thermal memory[18], and smallest in Lake Ontario, which is deeper and has consistently lower ice cover. Amplitudes were greater in recent decades compared to the early 20th century, although the record length is insufficient to assess the long-term stability of these oscillations.

Low-frequency oscillations effectively divided AAIC records into relative warming (P1: 1898 - 1941; P3: 1975 - 2022) and cooling, or neutral, (P2: 1941 - 1975) phases, which could be delineated by signal peak and trough years. Theil-sen estimators and associated bootstrapped confidence intervals (CI) highlighted significant differences in climate trends over each time interval (Figure 2). Median warming trends for P3 were generally stronger than those estimated for P1, although differences between trends were only rarely significant at the 85 or 90% confidence level (i.e., CI typically overlapped for all lakes). P2 was characterized by nearly neutral ice and air temperature trends (Figure S3), with marginal cooling that was only significant in Lakes Superior and Erie. There were no statistically significant differences between AAIC trends estimated over variable periods in Lake Ontario, which was characterized by relatively low (~10%) ice cover year over year.

SSA-derived AAIC signals were strongly correlated with the Atlantic Multidecadal Oscillation (AMO) and Pacific Decadal Oscillation (PDO) index (Figure 2). Correlation coefficients (R) between AMO and AAIC exceeded 0.60 for all lakes (Figure 2a,b), with strongest correlations in Lakes Huron ($R = -0.86$) and Superior ($R = -0.78$). Although correlations with PDO were smaller than AMO for all lakes, they exceeded 0.20 for Lake Ontario. Cross-correlation analysis indicated variable lags between lake and climate signals, with peaks that exceeded the white-noise threshold at 0 and 1 years for both PDO and AMO (Figure 2e–f). Power spectrum analysis further showed statistically significant oscillations at periods of ~60 years and ~20 years across all lakes, consistent with corresponding signals in the AMO and PDO indices (Figure 2g–h).

2.2 Shifts in Variability

Kernel density distributions and variances were estimated from detrended AAIC and AAAT time series (see Methods). The resulting parameter distributions (Figure 3) indicate notable shifts in variability across the climatological record. Several lakes showed reduced variability during the mid-century period (P2) as compared to the early- (P1) and late-century (P3) warming periods. For example, in Lake Huron, AAIC variance in P2 was nearly half that observed during P1 or P2 (70 vs. 159 and 127 $\%^2$). Ansari-Bradley tests highlighted statistically significant differences in variability over time for all lakes, although individual responses for AAIC and AAAT varied. Results also suggest that AAIC variability in Lake Ontario has decreased significantly since the beginning of the 20th-century, with changes linked to dramatic reductions in mean ice concentrations, with recent decades experiencing near-zero annual ice coverage maxima.

Changes in interannual AAAT variance were largely driven by shifts in winter air temperature variance over the same early- mid- and end-century periods (Figure 4). While the variance of monthly summer air temperatures (May - September) remained fairly constant over the last century (1 - 3 $°C^2$), winter temperature variance was significantly lower for P2 as compared to P1 and P3. This is especially evident for February, where monthly variance was reduced by over 50% (3 vs. 8 $°C^2$). Since winter temperatures act as a direct control on ice cover, these shifts in interannual air temperature variability were able to propagate to AAIC variability over the same periods. For P2, minor, but statistically significant, decreases in air temperature variability were also observed in the fall (October - December), with small increases in variability in spring (April).

3. Discussion

This study highlights the importance of long-term observational records for characterizing hydroclimate variability in the Great Lakes region. Previous work has often relied on relatively short records, particularly those beginning with the satellite era in the late 1970s. Our results show that these shorter records coincide with a period of rapid warming and heightened variability, which do not adequately represent long-term trends and can yield estimates up to an order of magnitude larger than those derived from the 125 year record used in this study. Warming trends for the modern era (1975 - 2022) were more than twice as large as those estimated for any other period over the last century, but they were not statistically different from those experienced in the early 1900s, which were also characterized by rapid air temperature increases and commensurate ice cover declines. The analysis presented here also demonstrates that ice cover exhibited an increasing or null trend between 1943 and 1975, with mid-century cooling rates that rivaled recent warming rates in Lake Superior. The largest declines in ice cover were observed in lakes with the highest median ice concentrations (Lakes Superior, Huron, and Erie), highlighting the role of ice-albedo feedback in amplifying ice loss[11]. Changes in air temperature, on the other hand, were most strongly tied to latitude, with southern lakes experiencing more rapid warming than northern lakes. In general, long-term warming trends were relatively consistent across the basin, and inter-lake differences were not statistically significant.

We have shown that AAAT and AAIC exhibited periods of increasing and decreasing trends over the last 125 years, but their magnitudes were not linearly proportional. For example, while warming and cooling rates on Lake Superior air temperatures were similar between P3 (0.30 ℃/decade) and P2 (-0.27 ℃/decade), ice cover loss was nearly 20% larger (-4.39 vs 3.74%/decade) than ice cover gain over the same periods. This response is best explained by ice-albedo feedback[11], where ice cover declines drive substantial reductions in surface albedo, allowing greater absorption of solar radiation and accelerating subsequent warming and ice loss. This feedback loop leads to hysteresis behavior in lake warming and cooling, such that cool-to-warm transitions occur more rapidly than warm-to-cool transitions, especially in deeper waters with large thermal masses[14]. One potential consequence of this hysteresis is a “ratcheting” effect in the warming of deep mid-latitude lakes, where symmetric (or warm-asymmetric) climate oscillations could drive an increase in lake temperatures over multiple warming and cooling cycles, even in the absence of a long-term monotonic climate trend. It is important to note that even minor asymmetries can disrupt this pattern and “reset” the lake thermal state. This is best exemplified by the El Niño-Southern Oscillation (ENSO), which is characterized by intense, short warming periods (El Niño) and weaker, more persistent cooling periods (La Niña) in the Great Lakes region[30], the latter of which allow for ice recovery and lake cooling.

This work highlights strong connections between multidecadal variability in the Great Lakes and large-scale decadal and multidecadal climate teleconnections. Long-term trends for lake climate variables were especially well correlated with AMO indices. Periods of rising AMO coincided with increasing AAAT and decreasing AAIC trends (and vice versa), and signals showed strong cross-correlation and spectral coherence at an oscillation period of ~60 years. Notably, accelerated warming from 1975 to 2022 coincided with rising trends in the AMO index, and the oft-cited regional climate shift in 1998, which is typically linked to an abnormally strong El Niño[31], also aligns with a transition in the PDO (positive-to-negative) and AMO (negative-to-positive) phases. There was additional evidence that background warming/cooling trends may exert some control on interannual AAAT and AAIC variability, especially during the winter. Increasing ice cover may act to “stabilize” lake-atmosphere interactions during cyclical cooling phases, effectively limiting atmospheric fluxes by “capping-off” open water.

Pacific and Atlantic teleconnection patterns influence climate in the Great Lakes region by affecting the ridge–trough system over North America, which determines the pathway and strength of the westerly jet stream[32]. Action centers for the ENSO and PDO are located upstream of the Great Lakes (i.e., Pacific), while action centers for the North Atlantic Oscillation (NAO) and AMO are located downstream along the westerly jet. Any upstream and/or downstream anomalies caused by these four patterns change the jet stream's ridge–trough intensity and pathway over North America[19]. These patterns act constructively and deconstructively, leading to complex climate interactions. In general, positive ENSO, NAO, and AMO phases lead to warm anomalies in the Great Lakes region[33], while positive PDO phases are associated with cold anomalies[19]. While NAO and ENSO drive short-term variability in jet-stream dynamics (i.e. months to years), AMO and PDO act on longer time scales, leading to more persistent pathways over decadal (or multidecadal) time scales. The combined effects of these teleconnection patterns significantly impact regional climate conditions, which in turn affect Great Lakes temperatures, ice cover, and surface fluxes. While no single pattern may be useful for explaining interannual ice cover variability[30,34], this work suggests that low-frequency teleconnection patterns are useful for describing long-term hydroclimate trends in the Great Lakes.

The framework used in this study demonstrates how relatively simple modeling approaches can be used to capture lake- and seasonal-scale relationships between air temperature and ice cover. It is important to note that more advanced methods, utilizing either physics based or machine learning approaches, are needed to resolve ice characteristics at smaller spatial and temporal scales. For example, this simplified approach may not be appropriate for estimating spatial ice cover distributions, as ice characteristics vary with both local climate conditions and lake bathymetry[35]. A temperature-only model may also be insufficient for simulating daily ice cover distributions, since ice transport and breakup depend on additional forcing conditions, including wind, precipitation, circulation, and surface waves[35]. The strong model performance described in this study ($R_{val}^2 \geq 0.93$; $RMSE_{max} \leq 6.01\%$) suggests that these processes become less important when ice characteristics are aggregated at lake-wide seasonal scales. AAIC is only marginally affected by daily ice cover variability, and the beta-regression approach implicitly accounts for variable bathymetry and heat content in each lake. As such, the simplicity of this modeling approach is not expected to have a significant influence on either the long-term hindcasts or trends discussed herein.

While multidecadal variability may modulate the timing and magnitude of lake responses, a significant long-term warming trend remains evident across all lakes. Natural systems vary at multiple scales, both spatially and temporally, and separating these variations is critical for anticipating ecosystem responses and designing adaptive management strategies. For example, while long-term warming drives an overall decline in ice cover, oscillatory modes such as the AMO and PDO can substantially amplify or dampen these trends on multidecadal timescales. As the AMO moves into future cooling phases in the Great Lakes region, we may anticipate a pause in ice cover decline, or even a *moderate* (short-lived) increase in average ice cover, as this study suggests occurred mid-century. This contradicts most existing literature on this subject[19,36,37]. It is important that we do not interpret these changes as climate recovery, but rather a temporary respite from the background warming that has continued over the last 125 years. The Great Lakes may never be completely “ice-free”, but we may very well see decades between periods of substantial ice cover. Understanding how these cycles may affect other physical, biogeochemical, and ecological processes remains a critical research question.

4. Methods

4.1 Air Temperature Observations

Air temperature records were sourced from Fujisaki-Manome et al.[28], who aggregated overland meteorology station measurements from across the Great Lakes region. In brief, daily mean air temperatures were calculated from the Global Historical Climatology Network daily (GHCNd)[38] and the Great Lakes Air Temperature/Degree Day Climatology, 1897 -1983[24] for various cities surrounding each lake (Figure S1). Although few single stations provided measurements over the entire data record, locations were selected such that neighboring stations could be combined to provide a single continuous time series for each city. Special attention was paid to ensure consistency in observations between station shifts (i.e. equipment upgrades, station relocations, etc.), including comparisons during overlapping measurement periods. Representative stations on each lake were averaged to produce a single, continuous lake-averaged air temperature record. Climate reanalyses (e.g., ERA5) often approximate lake surface conditions to climatological values in the pre-satellite period, and are also influenced by the growing number of meteorological and oceanic observations assimilated in time, which can create artifacts in long-term variabilities within a dataset. The reconstructed daily air temperature records are specifically designed to bypass this issue to enable research focusing on long-term trends and variabilities. Additional data processing information, including individual station coverage at each selected location, can be found in the metadata documentation and King et al.[39]. Air temperature observations were used to estimate annual average air temperature (AAAT), calculated as the mean of daily observations between January 1 - December 31 for any given year for each lake.

4.2 Lake Ice Modeling

A statistical regression model was developed to predict the annual lake-wide average ice cover (AAIC) for each of the Great Lakes. The model is based on three predictor variables derived from daily air temperature records: maximum freezing degree days (FDD_{max}), maximum thawing degree days (TDD_{max}), and antecedent heat (AH). Daily freezing and thawing degree days were calculated following the method of Assel et al.[27]. Freezing degree days (FDD) were computed as the cumulative sum of negative daily air temperatures over the winter season (October 1 to April 30), while thawing degree days (TDD) were based on positive daily temperatures. For both FDD and TDD, the cumulative sum was constrained such that it was reset to zero on any day when the addition of the temperature would otherwise yield a negative value. FDD_{max} and TDD_{max} were obtained as the maximum values within each winter season. Antecedent heat (AH), serving as a proxy for the thermal energy retained in each lake from the preceding warm season, was calculated as the sum of daily air temperatures between May 1 and September 30 of each year.

These three variables were incorporated into a four-parameter beta regression model to predict AAIC values as derived from NOAA Great Lakes ice charts[40]. The model was defined as:

$$\mathrm{logit}(AAIC) = FDD_{max}x_1+TDD_{max}x_2+AHx_3+x_0 \qquad (1)$$

where x_0, x_1, x_2, and x_3 represent fitted coefficients, and *logit*(AAIC)=log(AAIC/[1-AAIC]). A beta regression approach was selected due to the continuous, bounded nature of AAIC, which varies between 0 and 1 and is well described by the beta distribution. Time series of daily ice cover averages for each lake were used to estimate AAIC over the historical observational record (1973 - present), calculated as the mean ice cover between December 1st and April 30th of each year (available from NOAA GLERL: https://www.glerl.noaa.gov/data/ice/#historical). As far as the authors' are aware, these are the only lake-wide ice cover observations available for model calibration or validation. Although ERA5 (1940 - present), HadISST (1871 - 2000), HadISST.2 (1850 - 2007) include ice masks for the Great Lakes, records prior to 1980 are either generated from unrealistic 1D lake models[41] or represented by climatology from later years[42,43]. Model coefficients were calibrated using data from two periods, 1973–1993 and 2004–2021 (39 years in total), selected to coincide with relatively high and low decades of ice cover, respectively. Maximum likelihood (ML) estimation was used to determine best-fit parameters using the *fitLGDmodel* function in MATLAB R2024a[44]. Model performance was evaluated using an independent validation period from 1994–2003 (10 years), during which both the coefficient of determination (R^2) and root mean square error (RMSE) were used to assess predictive accuracy (described in section 2.1).

Sensitivity tests were conducted to assess the impact of calibration period selection on simulated AAIC for each lake (Supplementary Figure S4). Six alternative 39 year calibration periods were tested, including a split-decade period (A1: 1973 - 1983 and 1994 - 2021), a modern period (A2: 1983 - 2021), an early period (A3: 1973 - 2011), and three periods with years selected randomly without repetition (A4-6). All models had similar performance metrics, with $R^2 > 0.80$ and RMSE < 7% (computed over the entire observational record: 1973 - 2021). Importantly, the choice of calibration years did not affect estimated climatological trends, as differences between alternative models were not statistically significant (not shown). These tests provided confidence in the model's robustness and stability for use in long-term hindcast simulations.

4.3 Long-term hindcast and multidecadal trends.

The fitted regression ice model was applied to the full air temperature record to generate a 105-year time series (1898–2022) of annual lake-wide average ice cover for each of the Great Lakes. Linear trends were calculated using non-parametric Theil-Sen estimators for the full hindcast period (1898–2022), as well as for three sub-periods delineated by increasing and decreasing phases of the AMO index (P1:1898 - 1941; P2: 1941 - 1975; P3: 1975 - 2022). Mann-Kendall tests[45] were performed for trends estimated over each sub-period to investigate the statistical significance of apparent warming and cooling periods in the lakes. Finally, 85, 90, and 95% confidence intervals for Theil-Sen estimators were calculated using non-parametric bootstrap methods (i.e. bootstrapping by sampling with replacement).

In addition to linear trends, long-term nonlinear trends were extracted using Singular Spectrum Analysis (SSA), a non-parametric technique that decomposes time series into a set of interpretable components without requiring assumptions about the underlying data distribution or noise structure[46]. SSA has been widely applied in climate data analysis for isolating low-frequency variability and trend components. In this study, the first reconstructed component was interpreted as the dominant low-frequency trend.

To evaluate the influence of large-scale multidecadal climate variability on Great Lakes ice cover, two major climate indices were analyzed: the Atlantic Multidecadal Oscillation (AMO) index and the Pacific Decadal Oscillation (PDO) index. Two approaches were used to assess the influence of the multidecadal climate patterns on AAAT and AAIC. First, cross-correlation analysis was conducted between each climate variable and teleconnection index (*crosscor*; MATLAB R2024a[44]), and correlations were compared to white noise thresholds as a measure of statistical significance. Next, normalized power spectral densities were estimated for each variable using Welch's method (*pwelch*; MATLAB R2024a[44]). Power spectra were plotted alongside red-noise spectral densities for reference.

4.4 Changes in Ice Cover and Air Temperature Variability:
Changes in interannual variability were assessed for both annual average air temperature (AAAT) and AAIC over the entire data record (1898 - 2022). Interannual variability was isolated from externally forced trends (i.e. climate warming and multidecadal variability), using detrended time series, which were estimated by subtracting the SSA-extracted long-term nonlinear trends from the modelled or observed climate variables[8]. To quantify time-evolving changes in variability, kernel density distributions and variances were computed using 30-year centered moving windows with 50% overlap, beginning with the 1898–1927 (e.g., 1912–1942, 1927–1957, etc.). Statistical significance of differences in variance across distinct time segments was evaluated using non-parametric Ansari-Bradley tests[47].

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Author Contributions
D.C. led and coordinated the various components of the study. D.C. designed and performed the data analysis, and H.A., A.F.M., J.K., and J.W. provided input on additional analysis techniques. A.F.M., E.V., and K.B. prepared air temperature data for publication and performed preliminary data analysis. D.C., H.A., and M.R. contributed to writing this manuscript. All authors discussed the results, aided in their interpretation and contributed to editing the paper.

Competing Interests
The authors declare no competing interests.

Data Availability
Post-processed AAAT, AAIC, AH, FDD_{max}, and TDD_{max} timeseries for each lake (1898 – 2022) are available at https://doi.org/10.5281/zenodo.21362297. These timeseries are the source data for Figures 1, 2, 3, and Supplementary Figures S2, S3, and S4. The source data for Figure 4 and Supplementary Figure S1 are available from Fujisaki-Manome et al. (2024)[28]. Additional source data for Supplementary Figure S2 were original published by Yang et al. (2020)[40], with updated datasets available from NOAA GLERL at https://www.glerl.noaa.gov/data/ice/#historical

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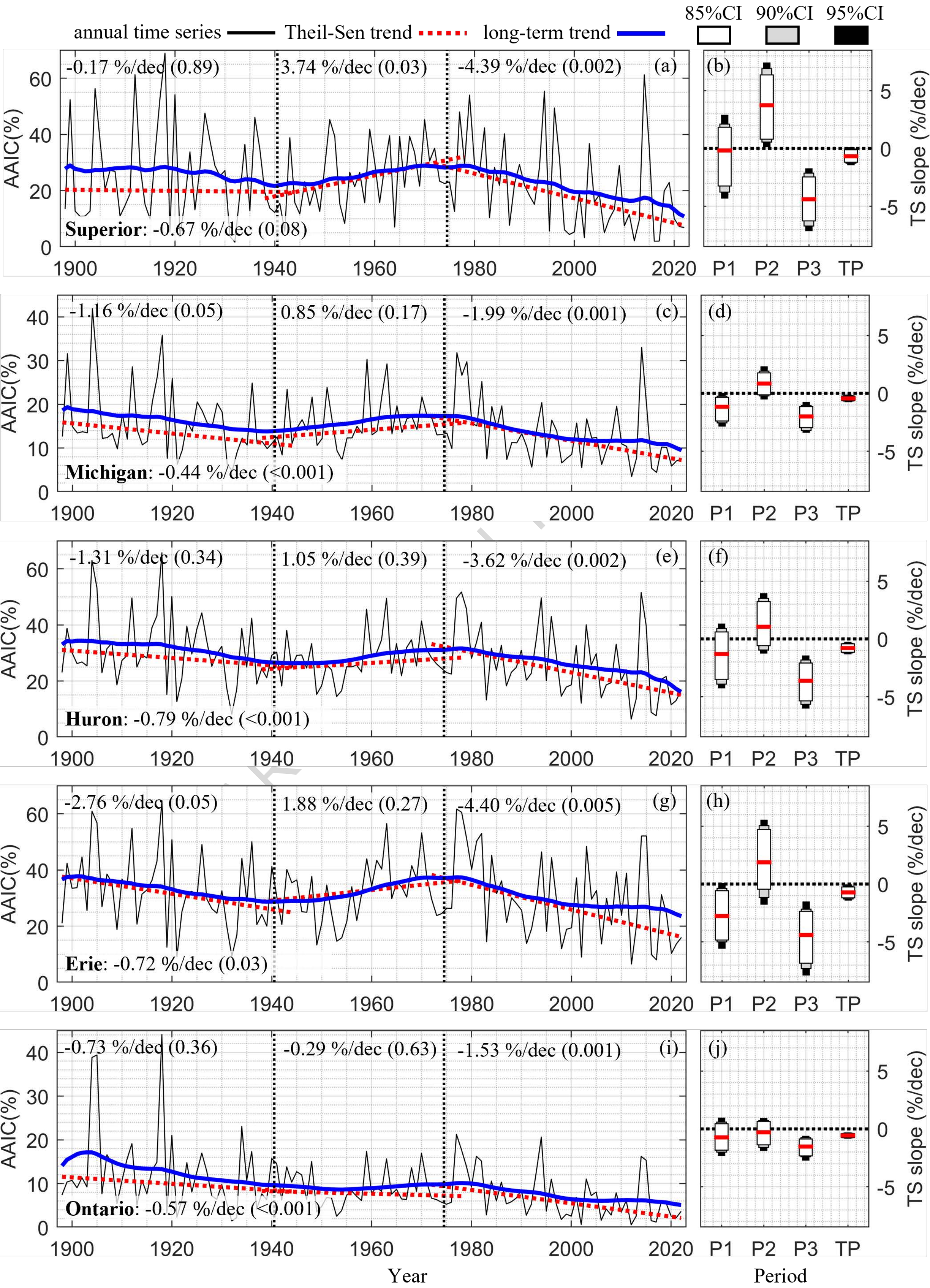
Figure 1: Long-term ice cover timeseries and trends. [**a,c,e,g,i**] Modeled annual average ice cover (AAIC; black line), SSA-derived long-term trends (blue line), and best-fit linear warming trends (red dotted lines) as calculated for periods 1898 - 1941 (P1), 1941 - 1975 (P2), and 1975 - 2022 (P3). Best-fit Theil-Sen estimators (%/decade) and associated p-values from Mann-Kendall tests (two-sided) are labeled for each period, including the total study period (TP: 1898 - 2022; adjoining lake label). [**b,d,f,h,j**] Boxplots indicate bootstrapped (n=1000) 95% (black), 90% (gray), and 85% (white) confidence intervals for Theil-Sen estimators calculated over each period, where the center (red line) indicates the best-fit estimator, as defined in panels [**a,c,e,g,i**]. Each row of subplots represents a separate lake, as labelled.

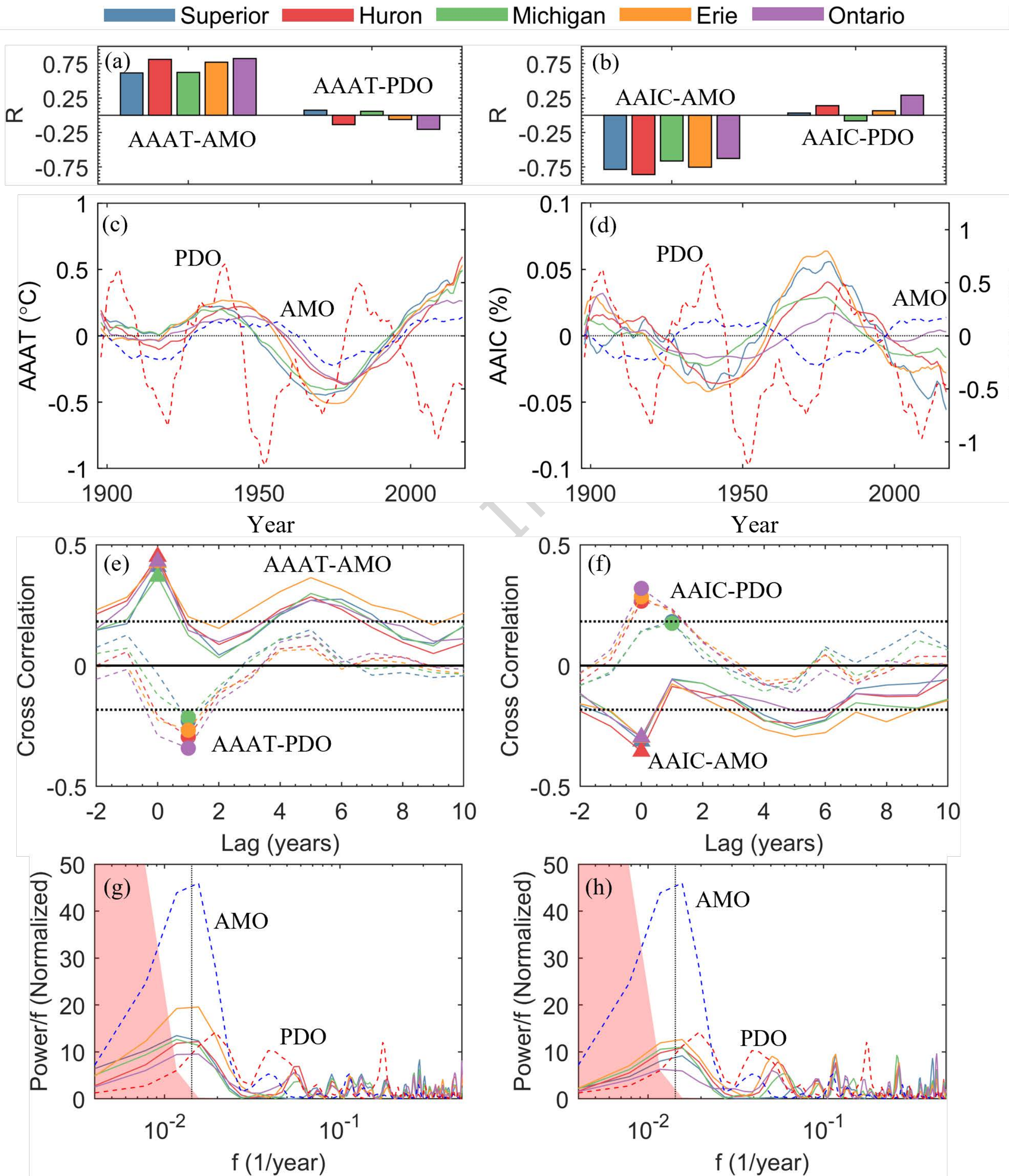
Figure 2: Low-frequency air temperature and ice cover signals and climate oscillations. Subplots include [**a,b**] correlation coefficients (R) estimated for low-frequency annual average air temperature (AAAT;) and annual average ice cover (AAIC; **b**) signals as compared to Atlantic Multidecadal Oscillation (AMO) and Pacific Decadal Oscillation (PDO) indices [**c,d**] time series of AMO (blue dash) and PDO (red dash) indices plotted alongside low-frequency AAAT (**c**) and AAIC (**d**) signals; [**e,f**] estimates of cross-correlation coefficients at various lag times for AAAT (**e**) and AAIC (**f**) as correlated with AMO (solid lines) and PDO (dashed lines). Maximum correlation lag times are marked for reference (circle: PDO; triangle AMO) and white noise

thresholds are indicated with horizontal blue lines; [**g,h**] normalized power spectrum for AAAT (**g**) and AAIC (**h**) time series on each lake, with normalized AMO (blue dash), PDO (red dash), and red noise (red shading) spectrum included for reference.

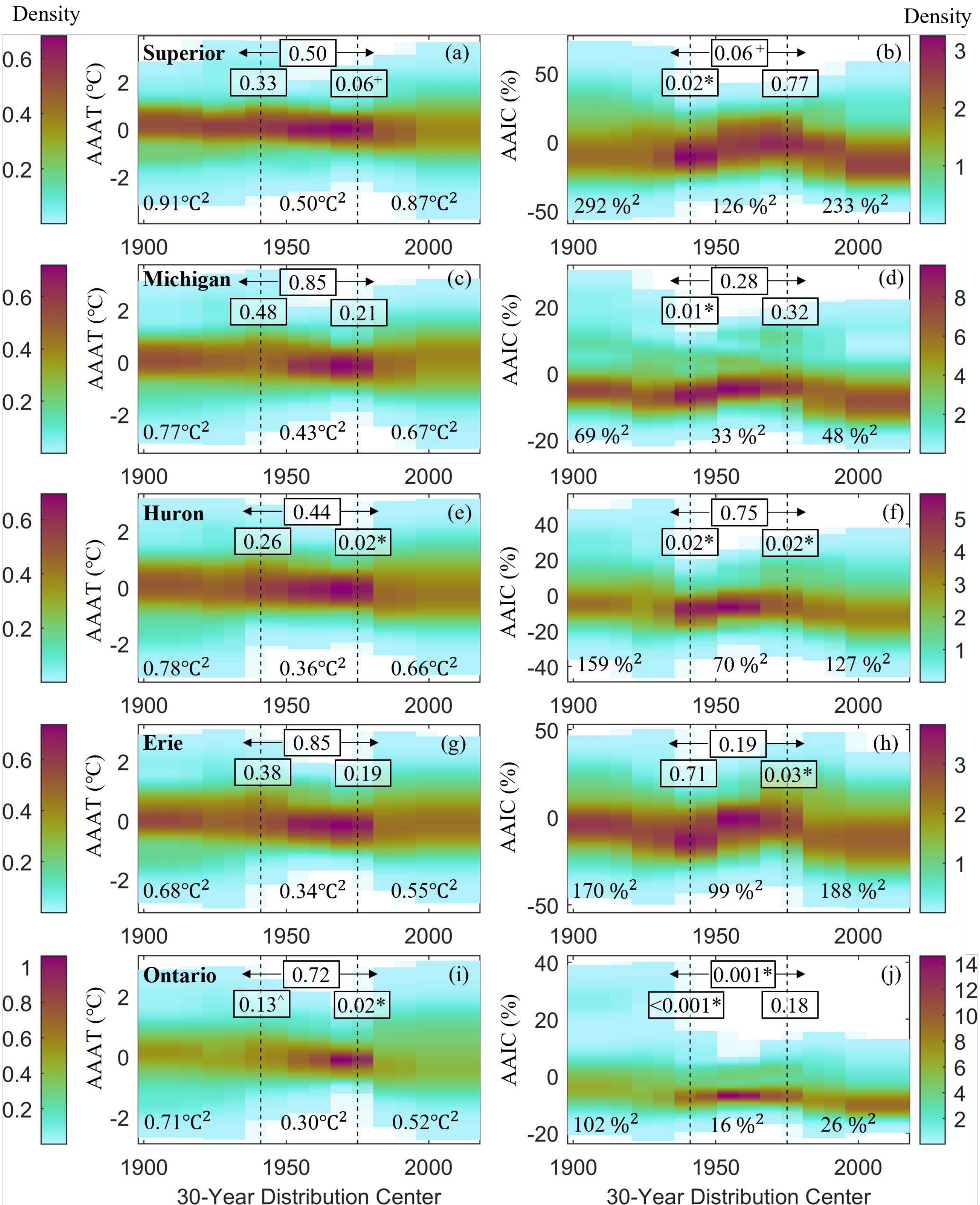
Figure 3: Kernel density distributions for air temperature and ice concentration. Distribution widths and magnitudes are indicated by colormaps, where subplots in the left column (**a,c,e,g,i**) represent AAAT and subplots in the right column (**b,d,f,h,j**) represent AAIC for each lake. Parameters are calculated from SSA-detrended time series, with kernel density estimates calculated on 30-year centered moving windows (50% overlap). The variance for each period (P1:1898 - 1941; P2: 1941 - 1975; P3: 1975 - 2022) is included for reference, and boxes indicate the p-value for Ansari-Bradley tests of variability (two-sided) between periods (*: $p<0.05$; +: $p<0.10$; ^: $p<0.15$).

Figure 4: Monthly interannual air temperature variance for each period. Periods are defined as in text (P1:1898 - 1941; P2: 1941 - 1975; P3: 1975 - 2022), and each subplot represents a different lake (**a**: Superior, **b**: Michigan, **c**: Huron, **d**: Erie, **e**: Ontario). Symbols are used to indicate months where the variance of P2 is significantly different than the variance of P1 or P3, as determined using pairwise two-sided Ansari-Bradley tests (*: $p<0.05$; +: $p<0.10$). P-values of Ansari-Bradley tests (P1 vs. P2 | P3 vs. P1) are included for reference.

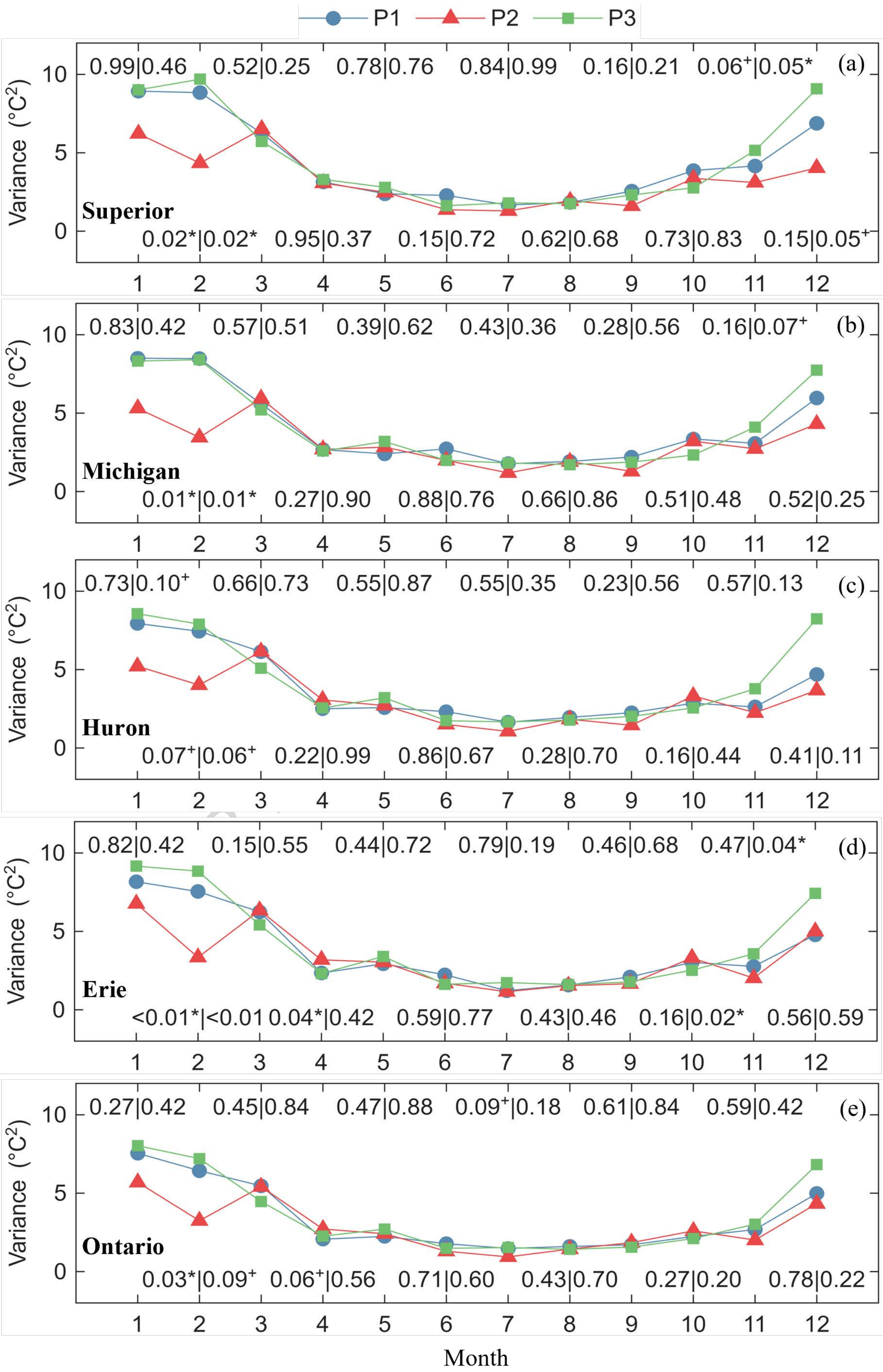

85%CI
90%CI
95%CI
annual time series
Theil-Sen trend
long-term trend
-0.17 %/dec (0.89)
3.74 %/dec (0.03)
-4.39 %/dec (0.002)
(a)
(b)
Superior: -0.67 %/dec (0.08)
-1.16 %/dec (0.05)
0.85 %/dec (0.17)
-1.99 %/dec (0.001)
(c)
(d)
Michigan: -0.44 %/dec (<0.001)
-1.31 %/dec (0.34)
1.05 %/dec (0.39)
-3.62 %/dec (0.002)
(e)
(f)
Huron: -0.79 %/dec (<0.001)
-2.76 %/dec (0.05)
1.88 %/dec (0.27)
-4.40 %/dec (0.005)
(g)
(h)
Erie: -0.72 %/dec (0.03)
-0.73 %/dec (0.36)
-0.29 %/dec (0.63)
-1.53 %/dec (0.001)
(i)
(j)
Ontario: -0.57 %/dec (<0.001)
AAIC(%)
TS slope (%/dec)
P1
P2
P3
TP
Year
Period



Superior
Huron
Michigan
Erie
Ontario
(a)
AAAT-PDO
AAAT-AMO
R
0.75
0.25
-0.25
-0.75
(b)
AAIC-AMO
AAIC-PDO
(c)
PDO
AMO
AAAT (°C)
Year
(d)
AAIC (%)
AMO & PDO
(e)
AAAT-AMO
AAAT-PDO
Cross Correlation
Lag (years)
(f)
AAIC-PDO
AAIC-AMO
(g)
Power/f (Normalized)
f (1/year)
(h)



Density
Density
Superior
(a)
0.50
0.33
0.06+
AAAT (°C)
0.91°C2
0.50°C2
0.87°C2
(b)
0.06+
0.02*
0.77
AAIC (%)
292 %2
126 %2
233 %2
Michigan
(c)
0.85
0.48
0.21
0.77°C2
0.43°C2
0.67°C2
(d)
0.28
0.01*
0.32
69 %2
33 %2
48 %2
Huron
(e)
0.44
0.26
0.02*
0.78°C2
0.36°C2
0.66°C2
(f)
0.75
0.02*
0.02*
159 %2
70 %2
127 %2
Erie
(g)
0.85
0.38
0.19
0.68°C2
0.34°C2
0.55°C2
(h)
0.19
0.71
0.03*
170 %2
99 %2
188 %2
Ontario
(i)
0.72
0.13^
0.02*
0.71°C2
0.30°C2
0.52°C2
(j)
0.001*
<0.001*
0.18
102 %2
16 %2
26 %2
1900
1950
2000
30-Year Distribution Center
30-Year Distribution Center


P1
P2
P3
0.99|0.46 0.52|0.25 0.78|0.76 0.84|0.99 0.16|0.21 0.06+|0.05*
(a)
Superior
0.02*|0.02* 0.95|0.37 0.15|0.72 0.62|0.68 0.73|0.83 0.15|0.05+
0.83|0.42 0.57|0.51 0.39|0.62 0.43|0.36 0.28|0.56 0.16|0.07+
(b)
Michigan
0.01*|0.01* 0.27|0.90 0.88|0.76 0.66|0.86 0.51|0.48 0.52|0.25
0.73|0.10+ 0.66|0.73 0.55|0.87 0.55|0.35 0.23|0.56 0.57|0.13
(c)
Huron
0.07+|0.06+ 0.22|0.99 0.86|0.67 0.28|0.70 0.16|0.44 0.41|0.11
0.82|0.42 0.15|0.55 0.44|0.72 0.79|0.19 0.46|0.68 0.47|0.04*
(d)
Erie
<0.01*|<0.01 0.04*|0.42 0.59|0.77 0.43|0.46 0.16|0.02* 0.56|0.59
0.27|0.42 0.45|0.84 0.47|0.88 0.09+|0.18 0.61|0.84 0.59|0.42
(e)
Ontario
0.03*|0.09+ 0.06+|0.56 0.71|0.60 0.43|0.70 0.27|0.20 0.78|0.22
Variance (°C²)
0
5
10
1 2 3 4 5 6 7 8 9 10 11 12
Month